

CODER CIRCUITS

Inputs, Outputs, MSB & LSB						
Circuit	Input	Input MSB	Input LSB	Output	Output MSB	Output LSB
3×8 Decoder	A[2:0]	A[2]	A[0]	Y[7:0]	Y[7]	Y[0]
8×3 Encoder	D[7:0]	D[7]	D[0]	Y[2:0]	Y[2]	Y[0]

DECODER

MAIN CODE

```
module decoder_3to8 (  
    input wire [2:0] A,  
    output reg [7:0] Y  
);  
  
always @(*) begin  
    case (A)  
        3'b000: Y = 8'b00000001;  
        3'b001: Y = 8'b00000010;  
        3'b010: Y = 8'b00000100;  
        3'b011: Y = 8'b00001000;  
        3'b100: Y = 8'b00010000;  
        3'b101: Y = 8'b00100000;  
        3'b110: Y = 8'b01000000;  
        3'b111: Y = 8'b10000000;  
        default: Y = 8'b00000000;  
    endcase  
end  
  
endmodule
```

TEST BENCH CODE

```
`timescale 1ns/1ps
```

```
module decoder_3to8_tb;
```

```
reg [2:0] A;
```

```
wire [7:0] Y;
```

```
decoder_3to8 uut (
```

```
    .A(A),
```

```
    .Y(Y)
```

```
);
```

```
initial begin
```

```
    A = 3'b000; #10;
```

```
    A = 3'b001; #10;
```

```
    A = 3'b010; #10;
```

```
    A = 3'b011; #10;
```

```
    A = 3'b100; #10;
```

```
    A = 3'b101; #10;
```

```
    A = 3'b110; #10;
```

```
    A = 3'b111; #10;
```

```
    $finish;
```

```
end
```

```
initial begin
```

```
    $monitor("Time=%0t | A=%b | Y=%b", $time, A, Y);
```

```
end
```

```
endmodule
```

ENCODER

MAIN CODE

```
module encoder_8to3 (  
    input wire [7:0] D,  
    output reg [2:0] Y  
);
```

```
always @(*) begin
```

```
    case (D)
```

```
        8'b00000001: Y = 3'b000;
```

```
        8'b00000010: Y = 3'b001;
```

```
        8'b00000100: Y = 3'b010;
```

```
        8'b00001000: Y = 3'b011;
```

```
        8'b00010000: Y = 3'b100;
```

```
        8'b00100000: Y = 3'b101;
```

```
        8'b01000000: Y = 3'b110;
```

```
        8'b10000000: Y = 3'b111;
```

```
        default: Y = 3'bxxx;
```

```
    endcase
```

```
end
```

```
endmodule
```

TEST BENCH CODE

```
`timescale 1ns/1ps
```

```

module encoder_8to3_tb;

reg [7:0] D;
wire [2:0] Y;

encoder_8to3 uut (
    .D(D),
    .Y(Y)
);

initial begin

    D = 8'b00000001; #10;
    D = 8'b00000010; #10;
    D = 8'b00000100; #10;
    D = 8'b00001000; #10;
    D = 8'b00010000; #10;
    D = 8'b00100000; #10;
    D = 8'b01000000; #10;
    D = 8'b10000000; #10;

    $finish;
end

initial begin

    $monitor("Time=%0t | D=%b | Y=%b", $time, D, Y);
end

endmodule

```

